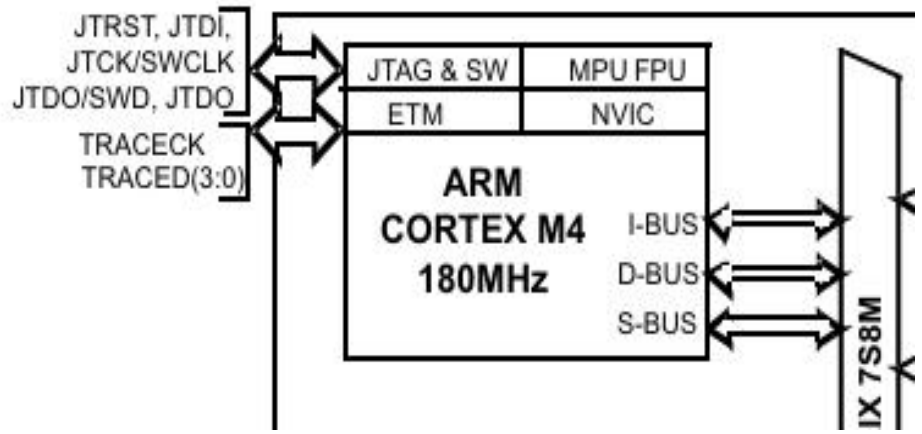
Nested Vectored Interrupt Controller (NVIC)

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What is NVIC?

NVIC is a hardware module inside the ARM Cortex-M4 processor which manages all interrupts generated by the microcontroller.

- **It decides which interrupt should be executed first based on priority.**
- **It ensures fast and efficient interrupt handling.**



Why Do We Need NVIC?

Without NVIC:

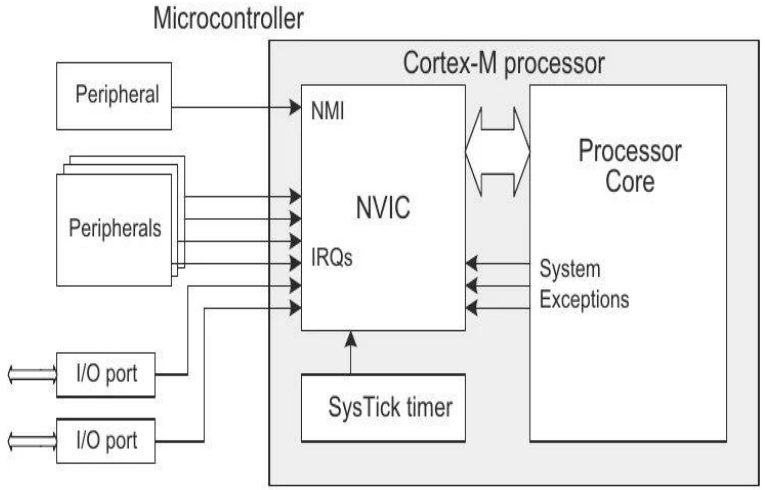
- CPU cannot efficiently manage multiple interrupts.
- Important interrupts may be delayed.

With NVIC:

- Handles multiple interrupt requests.
- Executes higher-priority interrupts first.
- Allows one interrupt to interrupt another if necessary.
- Reduces CPU response time.

How NVIC Handles Interrupts

- **Peripherals** such as Timer, UART, ADC, and GPIO generate **Interrupt Requests (IRQs)**.
- These IRQs are sent to the **NVIC**.
- NVIC **checks, enables, and prioritizes** the interrupt requests.
- The selected interrupt is forwarded to the **Processor Core**.
- The processor executes the corresponding **Interrupt Service Routine (ISR)**.
- After the ISR finishes, the processor **returns to the previous program**.
- **SysTick** and **System Exceptions** can also generate interrupts/exceptions.
- **NMI (Non-Maskable Interrupt)** is used for critical events.



Important Features of NVIC

- Supports **91 maskable interrupt channels**
- Supports **16 Cortex-M4 system exceptions**
- **16 interrupt priority levels**
- Supports **Nested Interrupts**
- Supports **Vectored Interrupts**
- Supports **Tail Chaining**
- Automatic processor state saving/restoring

Interrupt Priority & Nested Interrupts

Interrupt Priority

Each interrupt has a priority.

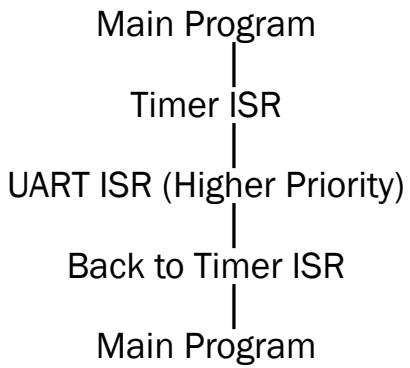
If multiple interrupts occur together:

Higher-priority interrupt executes first.

Interrupt	Priority
HardFault	Highest
Timer	Medium
UART	High

Nested Interrupt

A higher-priority interrupt can interrupt a lower-priority interrupt.



Vectored Interrupt & Interrupt Latency

Vectored Interrupt

Each interrupt has a fixed address in the Interrupt Vector Table.
CPU jumps directly to the correct ISR.

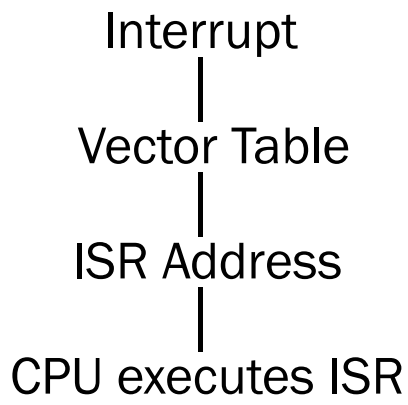
No searching is required.

Interrupt Latency

Time between interrupt occurrence and execution of the ISR.

NVIC reduces latency by:

- Direct vector table access
- Hardware priority checking
- Automatic context saving



Automatic Context Saving & Tail Chaining

Automatic Context Saving

When an interrupt occurs, NVIC automatically saves:

- R0–R3
- R12
- LR
- PC
- xPSR

After ISR execution, these registers are automatically restored.

Tail Chaining

Without Tail Chaining

ISR1

↓

Restore Registers

↓

Save Registers

↓

ISR2

With Tail Chaining

ISR1

↓

ISR2

Advantages

- Fast interrupt response
- Efficient priority management
- Supports nested interrupts
- Reduces CPU workload
- Improves real-time performance
- Faster switching between interrupt

Applications

- Robotics
- Automotive systems
- Industrial automation
- Medical devices
- IoT devices
- Motor control
- Embedded systems



Thank You